Constant-Current Battery Charger

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This circuit provides constant-current charging for batteries like nickel-cadmium (Ni-Cd), nickel-metal hydride (Ni-MH) and lithium-ion (Li-ion). The charger maintains constant current flow by using the property of the transistor that allows the current through the collector to be almost equal to the current through the emitter.

Zener diode ZD1 provides constant voltage across the base-emitter junction of transistor T1. Using variable resistor VR1 at the emitter of T1, one can set reference current I_{ref} flowing through the base-emitter junction of T2. I_{ref} is multiplied by the current gain (β) of T2 and flows through its collector. As the reference current is constant, the collector current (I_{Batt}) of T2 is also constant. The collector current is used to charge the battery. The relationship between I_{ref} and I_{Batt} (charging current) is as given below:

$I_{ref} = (Vz-0.6)/VR1$

$I_{Batt} \approx \beta_{T2} I_{ref}$

where Vz is the voltage across zener diode ZD1. The calculated values are approximate. Using this circuit, you can charge any battery as long as the charging voltage is at least 2V higher than the battery's terminal voltage. However, a minimum terminal voltage of 5.5V is recommended.

VR1 can also be adjusted by measuring the current flowing through the battery and setting it to the desired value. Transistors T1 and T2 should be isolated thermally. Use a suitable heat-sink for T2.

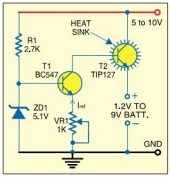


Fig. 1: Circuit of constant-current battery charger

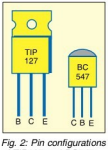


Fig. 2: Pin configurations of TIP127 and BC547 transistors

Electronic Glow-Worm

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Here is a circuit that simulates the chirping and glow of a glow-worm bug. The circuit is built around timer IC 7555 (IC1), which is wired as an astable multivibrator with a frequency of about 2.5 Hz. Other values of resistors R2 and R3 and capacitor C1 can be used if a different flashing rate is required.

The output from pin 3 of IC1 alternates between 0V and 3V, resulting

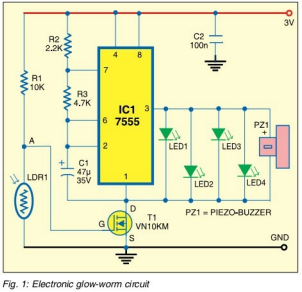


Fig. 1: Electronic glow-worm circuit



Fig. 2: PCB shaped as a bug